

DisSModel: A Python Framework for Spatially Explicit Dynamic Modeling

Sérgio Souza Costa¹, Nerval de Jesus Santos Junior¹, Felipe Martins Sousa¹, José Magno Pinheiro Alves¹, and Denilson da Silva Bezerra¹

¹ Universidade Federal do Maranhão (UFMA)

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Summary

DisSModel (Discrete Spatial Modeling) is a modular Python framework for spatially explicit dynamic modeling, targeting the complexities of Land Use and Land Cover Change (LUCC). It translates the modeling paradigms of the TerraME framework ([Carneiro et al., 2013](#)) into the Python ecosystem, enabling researchers to simulate complex socio-environmental systems — forest fires, epidemiological spreads, coastal dynamics — by coupling a time-stepped simulation clock with the spatial data structures of GeoPandas ([Jordahl et al., 2021](#)).

The framework provides a **dual-substrate architecture**: a vector substrate backed by GeoDataFrame for spatial expressiveness, and a raster substrate backed by NumPy 2D arrays for high-performance vectorised computation. DisSModel is available through the DisSModel GitHub organisation and on PyPI.

Statement of Need

Python has become the lingua franca for geospatial data science, supported by libraries such as GeoPandas and PySAL. However, these tools are designed for static analysis. Dynamic spatial modeling — simulating how landscapes evolve over time — has historically required specialised platforms. In Brazil, TerraME ([Carneiro et al., 2013](#)) and Dinamica EGO are the most widely adopted general-purpose frameworks, while institutions elsewhere often rely on narrower, more specific allocation models such as CLUE and CLUE-S ([Veldkamp & Fresco, 1996](#)). This fragmentation leaves researchers choosing between a general-purpose Lua-based toolchain and single-purpose implementations with no shared contract between them.

While TerraME is conceptually robust, its reliance on Lua scripting — a language with substantially smaller adoption in the data science ecosystem than Python — creates a barrier for data scientists, and the framework itself has seen no new release since August 2020. General-purpose simulation libraries, meanwhile, lack native synchronisation between a time-stepped clock and the geographical state of a GeoDataFrame. DisSModel fills this gap with a lightweight, purpose-built scheduler coupled directly to vector and raster spatial state, providing a Pythonic, actively maintained implementation of the TerraME paradigm — and, through its satellite packages, of specific allocation models like CLUE and CLUE-S as well.

Reproducibility is a first-class concern: the executor module provides a standardised lifecycle — `validate` → `load` → `run` → `save` — that captures provenance metadata (input checksums, parameters, timing, output paths) in an `ExperimentRecord` object generated automatically for every run, with no additional instrumentation by the modeller.

State of the Field

DisSModel occupies a niche between general-purpose agent-based modeling (ABM) libraries and specialised GIS simulation software:

Aspect	TerraME	Dinamica EGO	DisSModel
Language	Lua	Visual/Internal	Python
Simulation Engine	Discrete Event	Cellular Automata	Time-stepped scheduler
Spatial Structure	CellularSpace (Fixed)	Cellular Grid	GeoDataFrame + NumPy (Dual)
GIS Integration	TerraLib	Native Raster	GeoPandas / Rasterio
Extensibility	Script-based	Block-based	Class Inheritance
Reproducibility	Manual	Manual	Automated (ExperimentRecord)
Neighborhoods	GPM Support	Limited	libpysal weights (Queen, Rook, KNN, custom)

Frameworks like NetLogo and Mesa are excellent for ABM but require boilerplate to handle real-world spatial projections. DisSModel uses GeoPandas as its core engine, following the discrete spatial modeling approach proposed by Santos Junior (2025).

Software Design

DisSModel is organised into five modules with strict separation of concerns, extensible through class inheritance. **Core** manages the simulation clock: the Environment class orchestrates time progression, and spatial models auto-register at instantiation, receiving clock ticks through setup / pre_execute / execute / post_execute hooks. **Geo** provides the dual-substrate design: a vector substrate (SpatialModel, CellularAutomaton) on GeoDataFrame with libpysal neighbourhoods (Rey et al., 2021), and a raster substrate (RasterModel, RasterCellularAutomaton) on NumPy arrays with vectorised operations (shift2d, focal_sum, neighbor_contact) replacing cell-by-cell loops. **Executor** defines the ModelExecutor four-phase lifecycle (validate, load, run, save); subclasses self-register via __init_subclass__, and every run produces an ExperimentRecord capturing input checksum, parameters, timing, and output paths. **IO** provides a unified dataset abstraction (load_dataset / save_dataset) across GeoDataFrame, GeoTIFF, and Xarray/Zarr, including transparent s3:// resolution. **Visualization** integrates Matplotlib, Streamlit-compatible input widgets, and RasterMap for raster rendering.

This extensibility has already produced independent domain packages: dismodel-ca (Costa & Santos Junior, 2025a) (Cellular Automata patterns), dismodel-sysdyn (Costa & Santos Junior, 2025b) (System Dynamics components), and DisLUCC-Continuous (Costa et al., 2025), which implements the continuous LUCC modeling components of LUCCME — Demand, Potential, and Allocation (Veldkamp & Fresco, 1996; Verburg et al., 2004) — on both substrates and the same ModelExecutor contract, an explicit Python counterpart to TerraME/LUCCME.

Performance

The vector substrate offers spatial expressiveness; the raster substrate achieves high throughput via NumPy vectorisation. All benchmarks were run on an Intel Core i7-7700T CPU @ 2.90GHz, 15 GB RAM (Ubuntu, Python 3.12.3, NumPy 2.4.6, GeoPandas 1.1.3); absolute timings vary by hardware, but the relative speedup pattern is the result of interest.

Conway's Game of Life confirms mathematical equivalence across substrates with different throughput:

Grid	Cells	Raster (ms/step)	Vector (ms/step)	Speedup
10×10	100	0.12	74.81	639×
50×50	2,500	0.19	1,707.76	8,809×
100×100	10,000	0.41	7,069.14	17,394×
200×200	40,000	1.21	—	—
500×500	250,000	9.74	—	—
1,000×1,000	1,000,000	30.60	—	—

BR-MANGUE coastal dynamics. The foundation for coupled mangrove-flood modeling was established by Bezerra et al. (D. da S. Bezerra et al., 2013) and extended in D. da S. Bezerra et al. (2025), co-authored by Denilson da Silva Bezerra, the submitting author, and Felipe Martins Sousa — the same three researchers now responsible for the DisSModel reimplementation of BR-MANGUE. The brmangue-dissmodel package (Costa & Bezerra, 2026) validates the raster implementation against TerraME over the Maranhão Island dataset (50,496 cells, 20 steps): land use matches on 99.9% of cells, soil on 99.8%, and elevation (1 mm tolerance) on 90.7% (MAE 0.0033 m); match percentage is the appropriate metric for these categorical outputs (Pontius et al., 2008). Reproducible via brmangue-dissmodel/src/brmangue/executors/validation_executor.py against the committed TerraME golden CSV references in tests/fixtures/golden/, with tests/test_model_invariants.py and tests/test_transition_rules.py covering structural correctness.

Cross-substrate equivalence (60×60 synthetic grid, 3,600 cells, 10 steps) shows 100% match for land use and soil and 100% match for elevation under tolerance (MAE 0.000959 m, max error 0.024 m), with the raster substrate running at 2.1 ms/step against 84.2 ms/step for vector (40.1× speedup). The residual elevation divergence reflects floating-point rounding differences, not algorithmic disagreement. Each run automatically produces an ExperimentRecord with timings, checksums, and artifact paths.

DisSLUCC-Continuous implements the continuous CLUE-like allocation algorithm (Veldkamp & Fresco, 1996); MAE is the appropriate metric for its fractional outputs (Pontius et al., 2008; Willmott & Matsuura, 2005). Over the Lab1 study area (6,574 cells, 6 steps), both substrates reproduce the TerraME/LuccME reference at MAE = 0.0036 (RMSE 0.0062, max error 0.027) in [0,1] scale, consistent with expected divergence between independent implementations. Vector and raster are numerically identical (MAE = 0.000), and the raster substrate is 3.9× faster (44.0 ms/step vs 172.8 ms/step). Reproducible via disslucc-continuous/tests/test_benchmark_validation.py. Full end-to-end provenance from raw inputs to final metrics is addressed by the dissmodel-platform satellite package.

Research Impact Statement

DisSModel's scientific lineage is rooted in the TerraME/LuccME research program at INPE. The submitting author conducted doctoral research at INPE under Prof. Gilberto Câmara and Dr. Ana Paula Dutra Aguiar — principal architects of TerraME/LuccME — and has co-authored the LuccME modeling framework since 2009 (Costa et al., 2009). On 7 May 2026, DisSModel was presented at INPE's Graduate Program in Applied Computing seminar series (recording: <https://youtu.be/o7pMJt0CvXU>), connecting the framework to the institutional community that maintains TerraME and LuccME.

The framework is in active use across two UFMA research groups. Within LambdaGeo, graduate students develop disslucc-continuous and brmangue-dissmodel as part of their Master's research. Independently, Prof. Denilson da Silva Bezerra (UFMA, former INPE researcher), whose doctoral work established BR-MANGUE's scientific foundation (D. da S. Bezerra et al., 2013), actively uses the DisSModel reimplementation in his own coastal dynamics research program (PVCBS4959-2025, PVCBS4960-2025;

114 <https://sigaa.ufma.br/sigaa/public/docente/pesquisa.jsf?siape=3104707>), a collaboration
115 predating DisSModel itself (D. da S. Bezerra et al., 2025).

116 Starting August 2026, the project receives its first PIBIC-funded undergraduate research fellows
117 (CNPq), under approved institutional project PVCET5136-2026 at UFMA. The concentrated
118 2026 development effort was deliberately oriented toward this milestone: stabilizing the
119 ModelExecutor contract so each fellow can own an independent repository — disslucc-
120 continuous, disslucc-discrete, brmangue-dissmodel, or disscube (a data-cube access and
121 spatial-derivation layer, the Python successor to TerraME's fillCellularSpace) — without
122 core changes.

123 Since the original submission, development has continued with disslucc-discrete (Costa,
124 2026), a CLUE-S-like discrete allocation package using logistic regression — the discrete
125 counterpart to DisSLUCC-Continuous. An initial version has been validated against the Lab6
126 case study (Moju municipality, 5,914 cells, 6 steps) from the original TerraME/LuccME
127 repository, achieving 100% categorical agreement with the TerraME reference ($\kappa = 1.0$,
128 $F1 = 1.0$) at 56.8 ms/step.

129 These independent domain packages — dissmode-ca, dissmode-sysdyn, DisSLUCC-
130 Continuous, disslucc-discrete, and brmangue-dissmodel — demonstrate that the
131 ModelExecutor contract generalizes across modeling paradigms without core modifications.
132 Studies such as F. G. S. Bezerra et al. (2022), developed using LuccME, represent the class of
133 models the DisSLUCC packages are designed to reproduce. A roadmap toward DisSModel 1.0
134 (May 2027) anchors a community outreach effort including an open textbook, *Geospatial
135 Modeling with Python* (<https://lambdageo.github.io/geospatial-modeling-python/>), already in
136 progress. This positions DisSModel as the simulation layer in the Brazilian Earth Observation
137 stack, complementary to SITS (Simoes et al., 2021) and the Brazil Data Cube (Ferreira et al.,
138 2020).

139 Author Contributions

140 CRediT roles: S.S.C. — Conceptualization, Software (Core, Geo, Executor, IO, Visualization),
141 Methodology, Validation, Writing, Supervision, Project administration. N.J.S.J. —
142 Conceptualization, Software (initial design), Validation, Writing (undergraduate thesis (Santos
143 Junior, 2025)). D.S.B. — Conceptualization (domain science), Validation, Resources (D. da
144 S. Bezerra et al., 2013, 2025). F.M.S. — Software (brmangue-dissmodel), Data curation,
145 Validation (D. da S. Bezerra et al., 2025). J.M.P.A. — Software (disslucc-continuous),
146 Validation. All authors reviewed and approved the final manuscript.

147 AI Usage Disclosure

148 Development followed several phases. The initial prototype (May–June 2025), corresponding
149 to the second author's undergraduate thesis (Santos Junior, 2025), did not involve generative
150 AI. Development resumed in February–March 2026 with Claude (chat) used mainly for
151 documentation; from April 2026, Gemini CLI accelerated code generation and refactoring; from
152 June 2026, Claude Code (CLI) was used on newer satellite repositories such as disslucc-
153 discrete. AI tools also assisted with writing in English, which is not the submitting author's
154 native language. The scientific design — the TerraME compatibility contract, executor pattern,
155 dual-substrate architecture, and validation methodology — predates and is independent of
156 this AI-assisted phase, tracing to the submitting author's doctoral research at INPE and the
157 undergraduate thesis cited above. AI tools assisted with implementation velocity and language
158 clarity, not scientific or architectural decisions. All outputs were reviewed and validated by the
159 human authors.

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